# NyaySaathi (न्याय साथी) — AI Legal Assistant for Everyday Money Disputes in India

**Kuku — AI Generalist Assignment | Problem 2: Build an AI Lawyer | Submission Document**

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*GitHub Repository: https://github.com/abhishekp12345/NyaySaathi • Local/Hosted: Gradio UI Port 7860*

## 1. Problem Choice, Scope & Safety Guardrails

- **The Problem:** In India, millions of ordinary people lose money in everyday disputes—landlords keeping rental deposits, websites refusing refunds, or employers withholding final salaries. Hiring a lawyer costs ₹5,000 to ₹10,000 just to draft a legal notice, so people with ₹15,000 to ₹50,000 claims simply give up.
- **Our Scope:** Instead of trying to answer every complex legal question, NyaySaathi focuses on one clear goal: recovering money under ₹50 Lakh before going to court. It handles 3 high-volume disputes: (1) Rental Security Deposits (Transfer of Property Act 1882); (2) Consumer Refunds & Defective Goods (Consumer Protection Act 2019); (3) Unpaid Salaries & Settlements (Code on Wages 2019).
- **Safety Guardrail:** NyaySaathi strictly refuses to give advice on criminal matters, threats, physical violence, family disputes, divorce, or cases already in court. When threats or danger are detected, the AI halts legal drafting immediately and provides official emergency helplines: Police Emergency: 112 (with advice to file an FIR), NALSA Free Legal Aid: 15100, and National Consumer Helpline: 1915.

## 2. How the System Works: Generative AI vs Pure Deterministic Code

To prevent AI hallucinations, we do not let the language model run as an unconstrained black box. We divide tasks strictly between AI (for communication) and reliable Python code (for legal accuracy):

- **What AI Handles (Language & Empathy):** Listens to voice notes or text in Hindi, English, and Hinglish; pulls out the key dispute facts; explains legal options in 5 simple plain-English/Hindi sections; writes a polite pre-litigation legal notice; and simulates the opponent (landlord/company) in a 4-round practice chat to help citizens prepare before negotiating.
- **What Pure Code Handles (Zero Hallucination):** Limitation deadline clock (simple math in Python counting remaining days under Limitation Act 1963 / Sec 69 CPA 2019); verified Bare Acts database (pulls only real sections from India Code); citation validator (blocks fake section numbers and intercepted repealed laws); and evidence cross-checker (flags differences between claimed sums and bank payment receipts).
- **Human in the Loop:** The user verifies and confirms all extracted facts on the screen before any legal notice is drafted. Nothing is generated automatically without the citizen's approval.
- **Economics & Unit Cost:** An end-to-end case uses ~5,000 tokens across all steps. Using fast models like Groq Qwen 2.5 or Gemini Flash, each case costs about ₹0.13 (13 paise). Helping 10,000 citizens every day costs less than ₹1,250 per day, making free public access fully practical.

## 3. Testing, Benchmark Results & Honest Failures

- **20-Case Quality Benchmark:** Evaluated on 20 realistic legal situations (including voice notes, regional dialects, trick questions on repealed laws, and criminal threats): (1) Dispute Identification: 100% (20/20); (2) Correct Statutory Provision: 100% (20/20); (3) Zero Fake or Repealed Citations: 100% (20/20 — 0 hallucinations); (4) Missing Information Recognition: 100% (20/20); (5) Correct Emergency Escalation: 100% (20/20).
- **Honest Mistakes & How We Fixed Them:** (1) Quoting Repealed Laws: Early on, the AI cited the Payment of Wages Act 1936 and Evidence Act 1872. Both were repealed by the Code on Wages 2019 and Bharatiya Sakshya Adhiniyam 2023. We fixed this by adding a validator in code that intercepts and replaces repealed laws. (2) Missing Details in Messages: When users forgot to mention the claim amount, the AI threw an error. We resolved this by adding automatic follow-up clarification questions. (3) State Tenancy Nuances: Rental laws differ between states (e.g. Maharashtra vs Karnataka). The system now explicitly asks for the user's city and state before giving rental advice.

## 4. Submission Deliverables Summary

- **1. Prototype & Repository:** GitHub repository at https://github.com/abhishekp12345/NyaySaathi with full source code, architecture, 100% automated test benchmarks, and local/cloud launch instructions.
- **2. Pre-Litigation Notice & Action Plan:** Formal Legal Notice and 4-Step Recovery Action Plan downloadable as editable Microsoft Word documents (.docx) with clear [[PLACEHOLDER]] markers.
- **3. Raw Test Outputs:** Complete raw unedited test results and metrics preserved in outputs/evaluation_report.json and outputs/validator_log.jsonl.
- **4. Legal Disclaimer:** All legal drafts include a clear statutory disclaimer stating this tool provides educational legal information and does not constitute formal advocate representation under the Advocates Act, 1961.